Model Regularization

Author: Rohit Mande
Founder and CEO, inttrvu.ai

Website: https://inttrvu.ai

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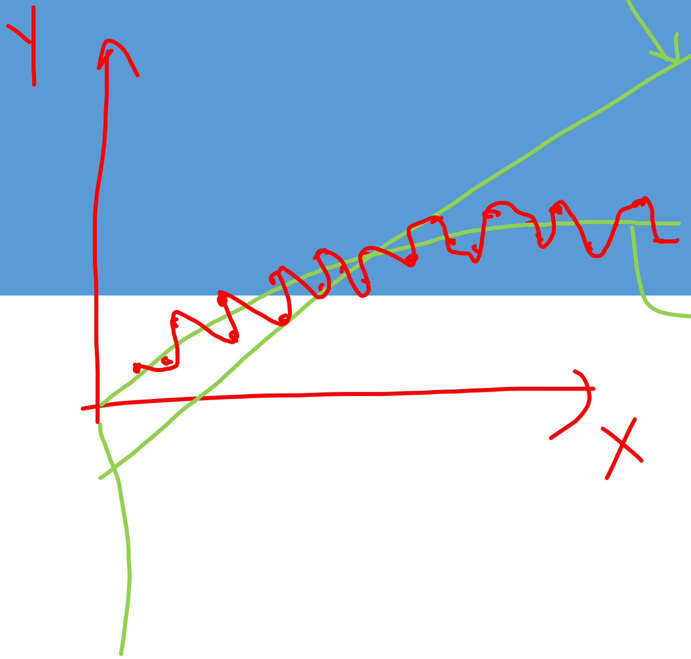
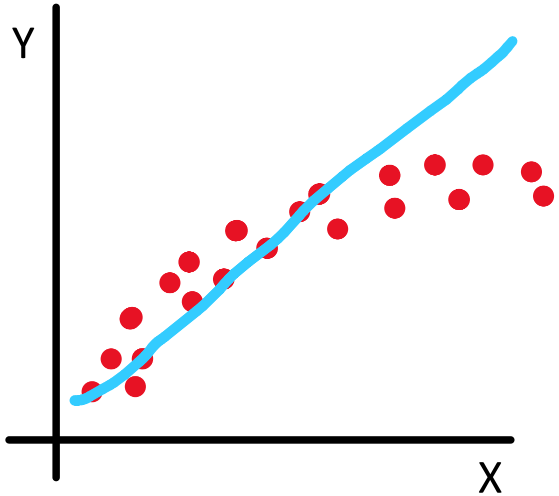


underfitting → not able to capture the trend of the data

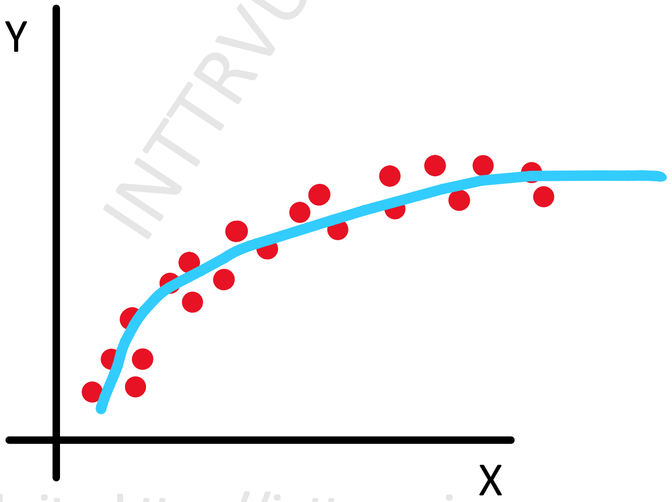
overfitting → capturing the noise in the data

Underfitting and Overfitting

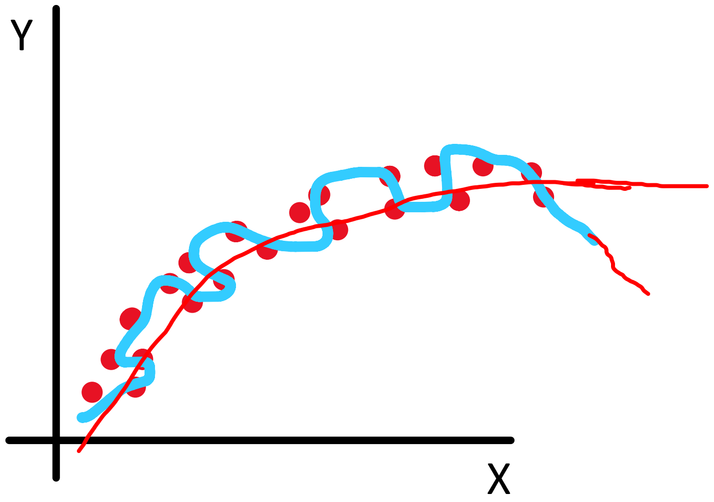
Underfitting and Overfitting



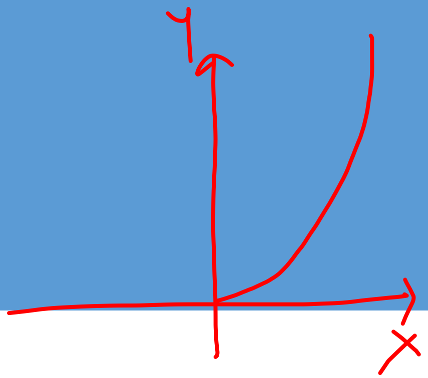
Underfitting



Optimal balance



Overfitting



→ real world data

We made a simplifying assumption and created a linear very simple model

⇒ We are trying to model a complex problem with a very simple model

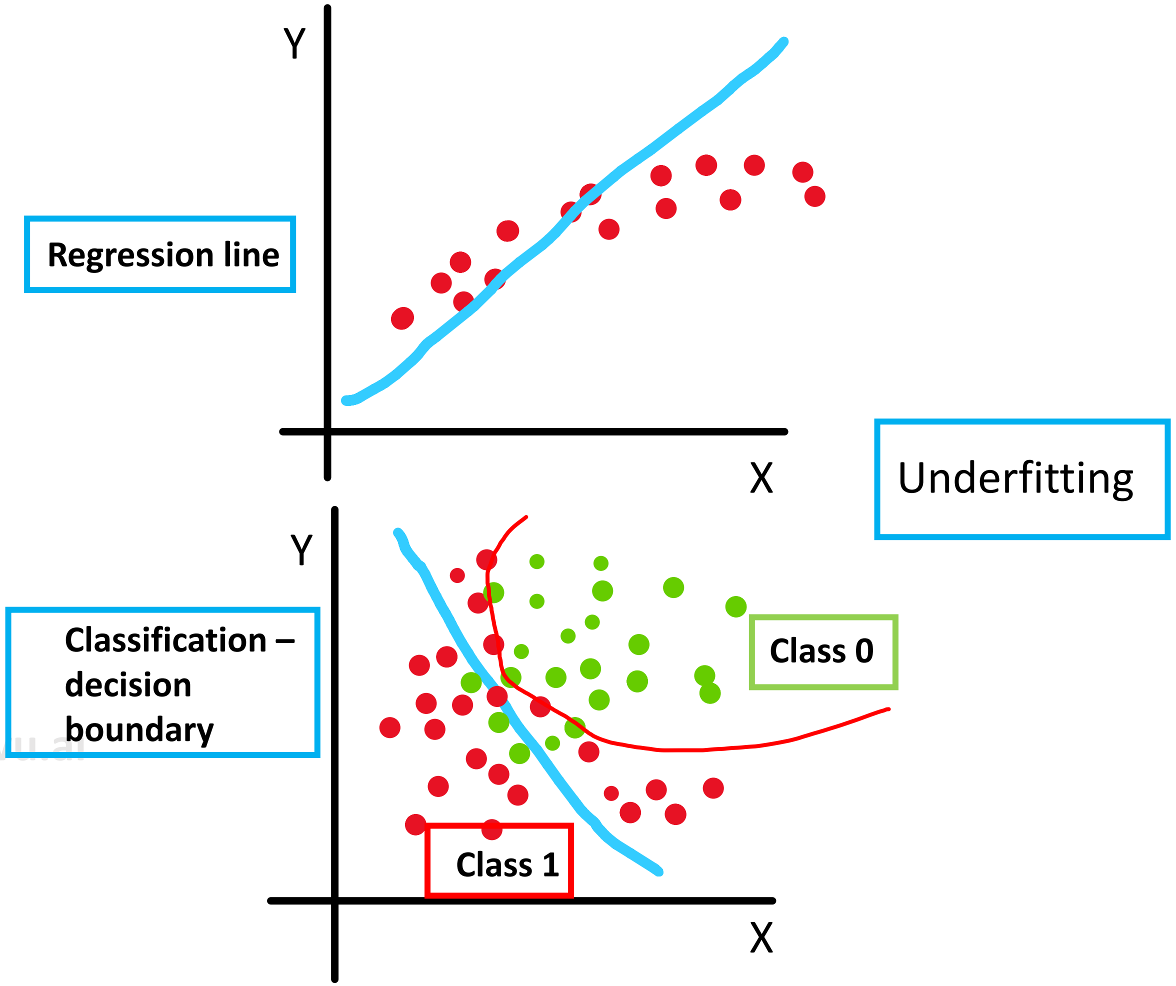
Bias and Variance

Bias ⇒ Underfitting ⇒ ML model is unable to capture the trend

- Bias represents an inability of machine learning algorithm to capture the true relationship between the data points

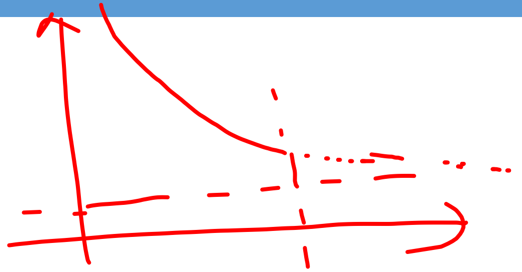
- Bias is the error introduced by approximating a real-world problem with a simplified model.

- Model with high bias are called as underfitted models



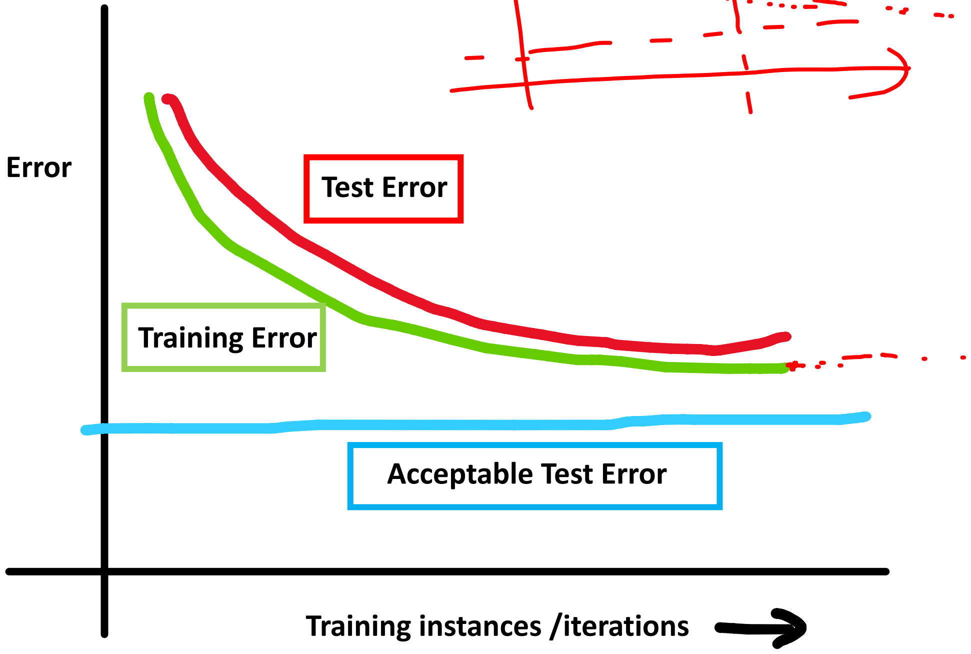
Bias

Increase the complexity of the model
↳ feature engineering
↳ capture more data | more features
↳ you can change the algorithm



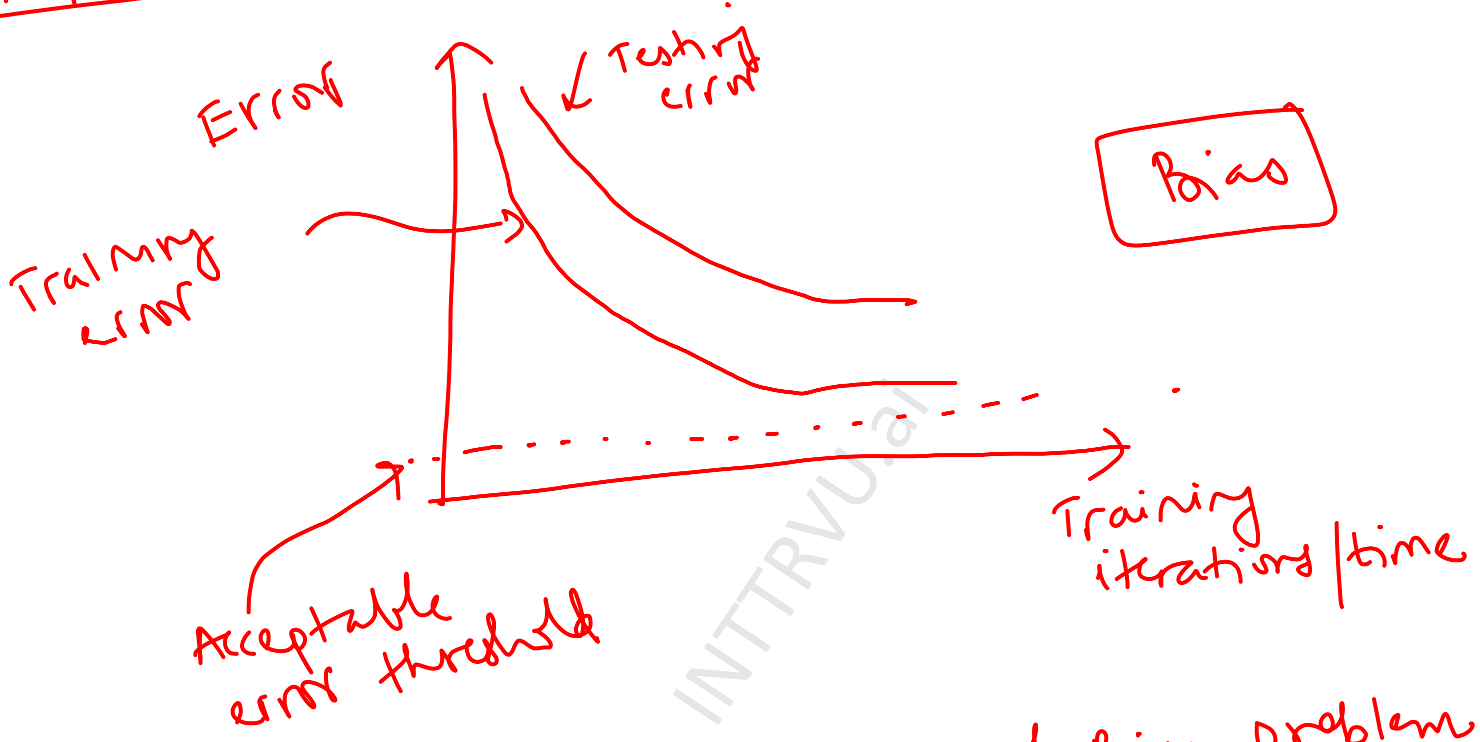
- Models with high bias give high training error and high error on test set as well

- If the current features are not able to capture all the factors contributing to response variable (what we are predicting) then it will result in high bias (underfitting)



Underfitting

Error Graph

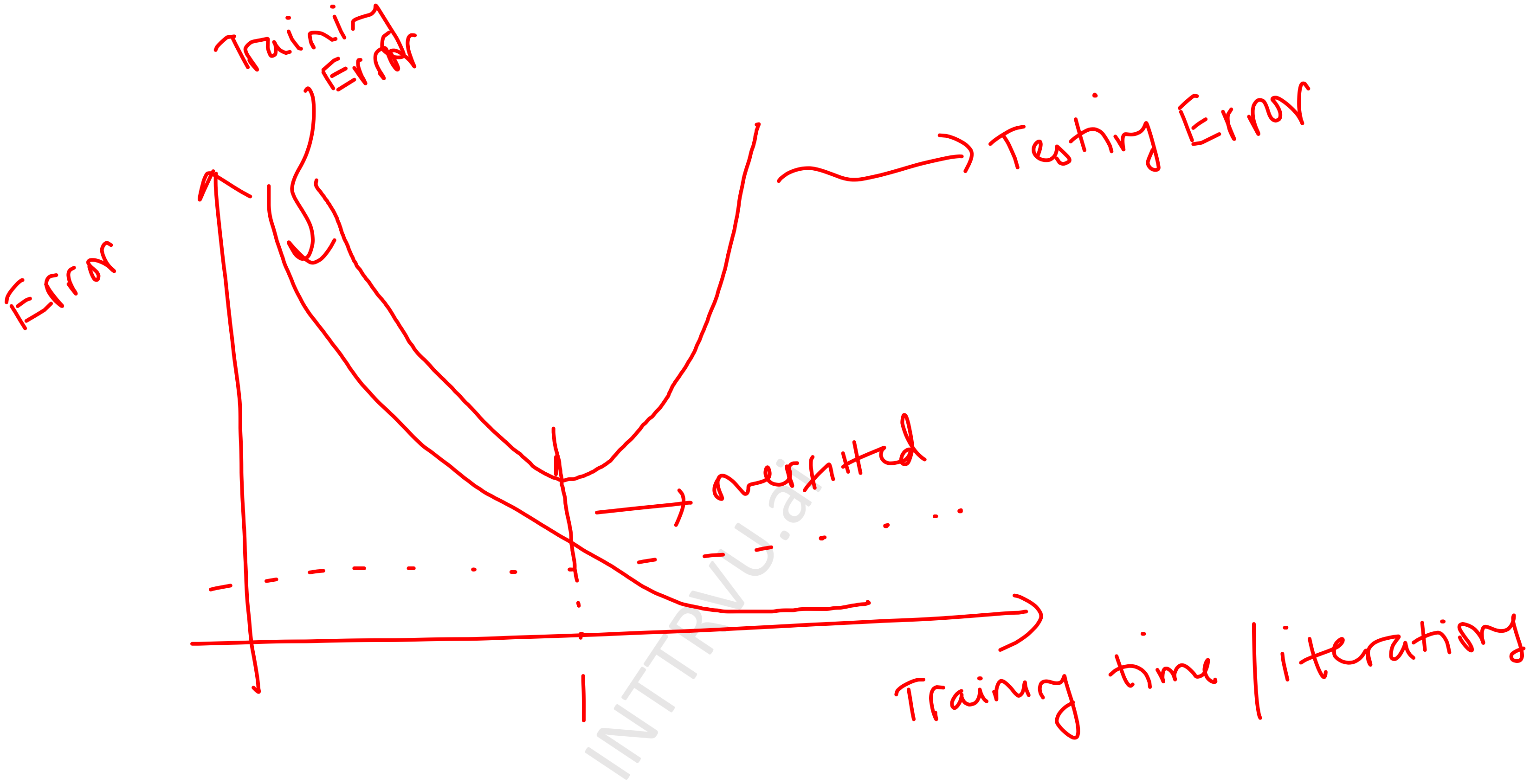


Bias

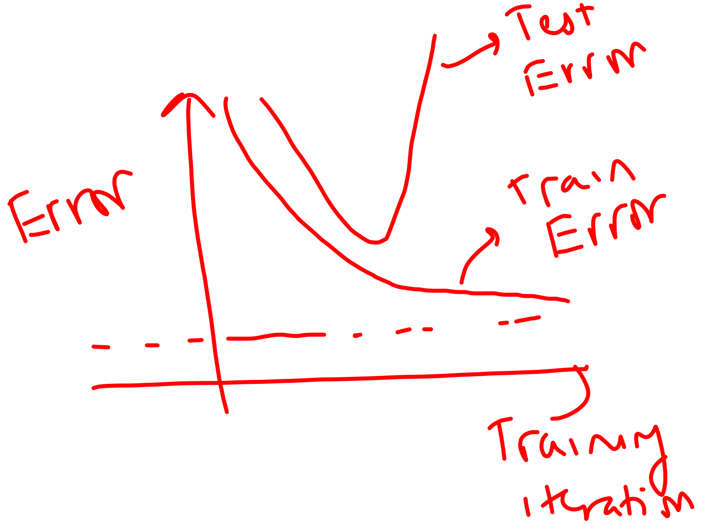
Training data
Testing data

Testing error > Training error

Symptom of Bias problem



Variance Symptom



Bias

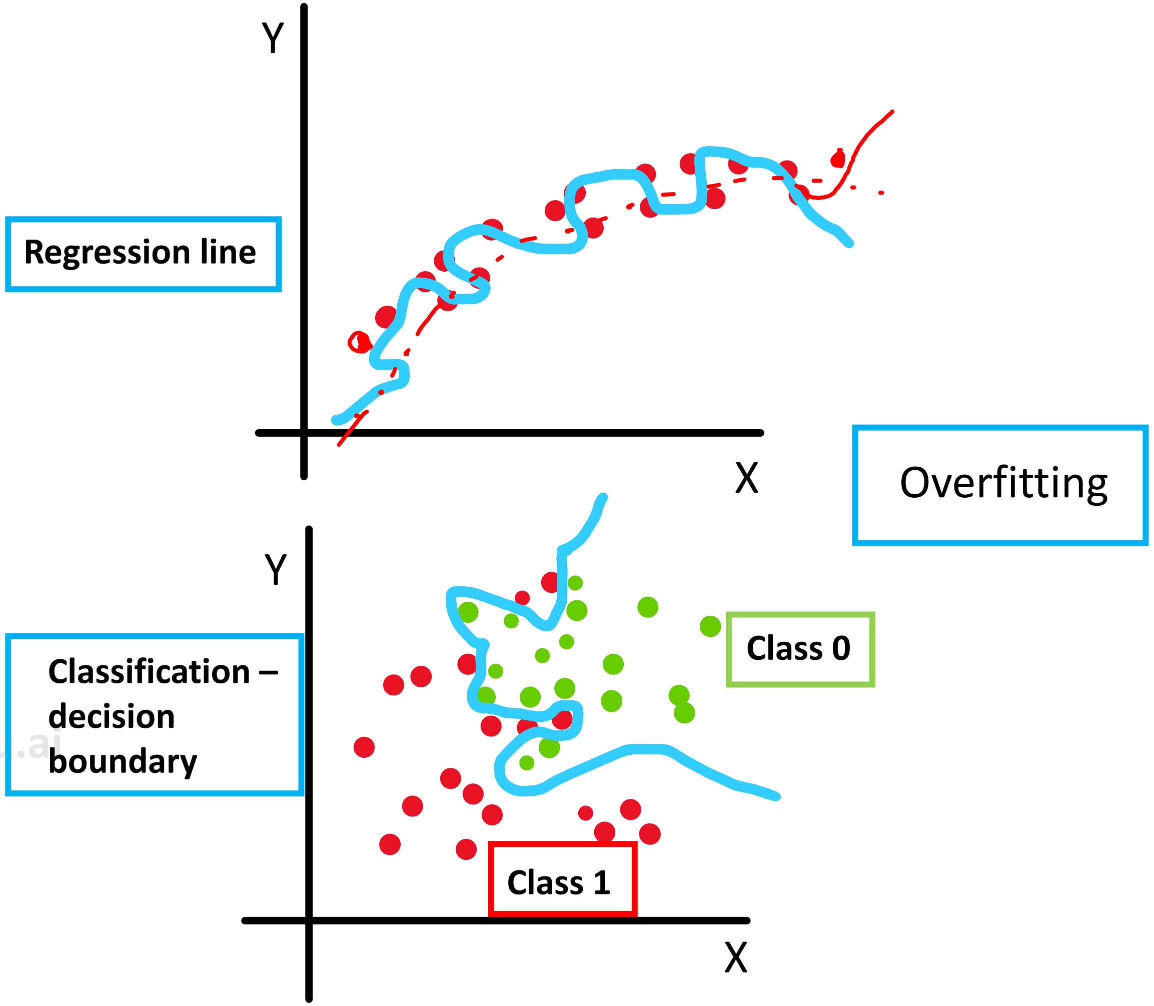
How to reduce model bias:

- Explore other potential features by discussing with business stakeholders
- Better optimization of model parameters
- Increase number of iterations and/or training data
- Create new features by using current features → feature engineering.
  - In house price prediction convert room sizes into carpet area,
  - In case of classifying engine as defective or non-defective, using production date and sell date to create number of months on shelf as a new feature

Variance

idea is to generalize the data and not specialize it

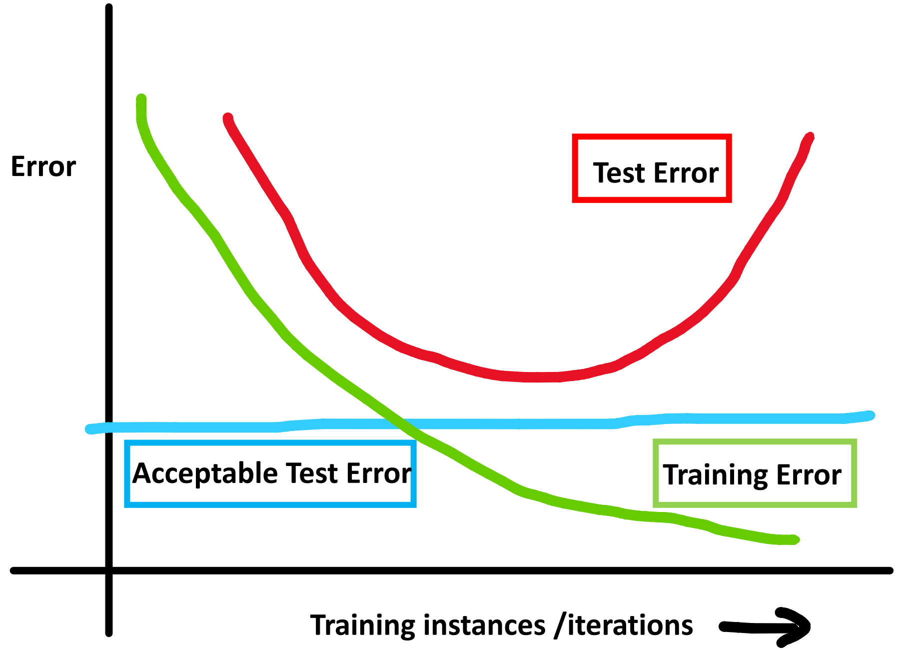
- Variance is inability of machine learning model to generalize learnings from training data
- Variance is the error introduced by the model's sensitivity to variations in the training data.
- Models with high variance can predict accurate results on training set but have high error on test (unseen) data
- This is result of overfitting on the training data



Variance

- Models with high variance give low error on training data but high error on test (unseen) data
- Highly complex models with large number of features can result in high variance models

- **How to reduce model variance:**
  - Reduce input features → θ parameters
  - Reduce number of parameters in model (use less complex model)
  - Increase training data (# of rows)
  - Introduce **regularization**



Overfitting

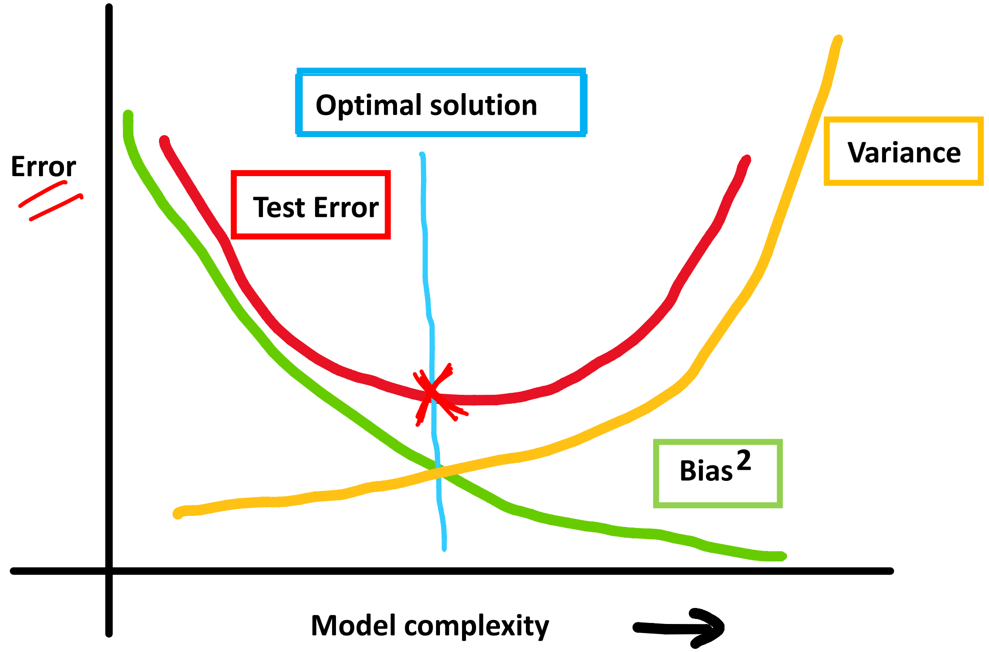
Bias variance trade-off

θ [1, 2, . . . , 100]

- As model complexity increases variance tends to increase and bias decreases

- Models with less complexity generally result in high bias (depends on data)

Hyper parameter optimiztion



Number of θ parameters

Cost Function = Total Error + λ Penalty Term

Penalty Term $\propto$ complexity of the model

$= R(\theta)$

λ decides how much to focus of Penalty Term

If λ is small $\Rightarrow$ Focus is on total error

λ is large $\Rightarrow$ Focus is on penalty term

λ is a hyperparameter

Regularization

to fight the overfitting

Regularization

Cost Function = Total Error + Penalty Term

due to regularization

- Regularization is the method of creating simpler (less complex) models
- It is achieved by adding penalty term to cost function
- It helps in reducing complexity of model by reducing some feature coefficients (θ_1, θ_2) etc. towards zero

Penalty Term ~ complexity

- Multiple linear regression cost function with regularization:

$$\underbrace{J(\theta)}_{\text{cost function}} = \underbrace{\frac{1}{2M}\sum_{i=1}^{M} (\theta_0 + \theta_1 X_{i1} + \theta_2 X_{i2} - Y_i)^2}_{\text{Total Error}} + \boldsymbol{\lambda}\,\mathbf{R}(\boldsymbol{\theta}) \rightarrow \text{Penalty Term}$$

λ Controls contribution of regulization function to cost function

$\mathrm{R}(\theta)$ is the regulization function chosen to impose penalty

Total penalty = $\boldsymbol{\lambda}\,\mathbf{R}(\boldsymbol{\theta})$

Penalty Term : $\lambda R(\theta) \longrightarrow$ regularization function

$\lambda \downarrow$ Controlling parameter

$R(\theta)$

L1 Regularization

$$R(\theta) = \sum_{i=1}^{M} |\theta_i|^1$$

$$= |\theta_1| + |\theta_2| + |\theta_3|$$

$$\hat{Y} = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \theta_3 X_3$$

$\uparrow$ intercept

Cost function = Total Error + $\lambda \left(|\theta_1| + |\theta_2| + |\theta_3| \right)$

By decreasing the cost function $\longrightarrow$ decreasing the penalty $\longrightarrow$ decreasing the $|\theta|$ parameters

$Y = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \theta_3 X_3$

complexity of a model can be measured by number of parameters, magnitude of the parameters.

Iteration 1. $\boxed{\text{Cost function} = \text{Total Error} + \text{Penalty}}$

$\lambda = 1$

Gradient Descent $\theta_0 = 1,\ \theta_1 = 1,\ \theta_2 = \theta_3 = 1$

Iteration 1 = Cost Function : 23 + 3 = 26

$\theta_0 : 0.9 = \theta_1 = \theta_2 = \theta_3$

Cost Function = 20 + 2.7 = 22.7

Iteration 2 : $\theta_i \leftarrow \theta_i - \eta \dfrac{\partial(\text{Cost Function})}{\partial \theta_i}$

L1 Regularization = Lasso regularization

- L1 regularization adds a penalty equal to the absolute value of the magnitude of coefficients.
- This type of regularization can result in sparse models with fewer features
- Some coefficients can become zero and eliminated from the model (less important features are removed)

- Multiple linear regression cost function with L1 regularization: **(Lasso Regression)**

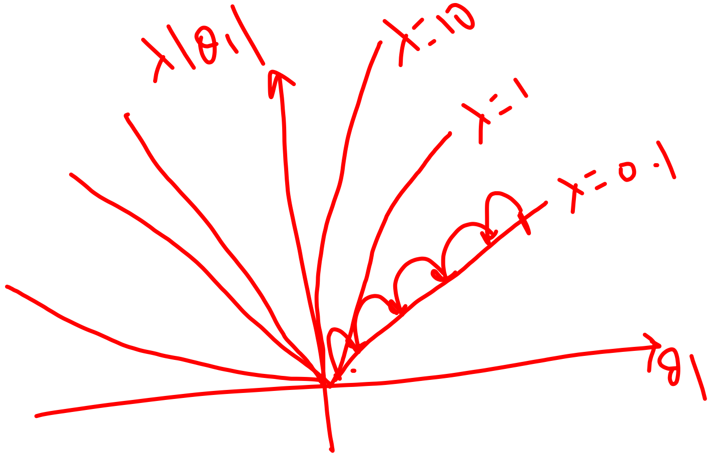
$$J(\theta) = \frac{1}{2M}\sum_{i=1}^{M} (\theta_0 + \theta_1 X_{i1} + \theta_2 X_{i2} - Y_i)^2 + \lambda \sum_{j=1}^{n} |\theta_j|$$

θ_0 not included

$|\theta_j|$ *represents absolute value of coefficient*

In this case n will be 2 as we have considered 2 features

Note: Regularization is typically used when there are large number of features.



$$\theta_i \Leftarrow \theta_i - \eta \frac{\partial(\text{Cost function})}{\partial \theta_i}$$

$$\theta_i \Leftarrow \theta_i - \underbrace{\eta \lambda}$$

L1 → sparse model.

Sparse Models

$$\hat{Y} = \theta_0 + \sum_{i=1}^{100} \theta_i X_i$$

For a sparse model, some of these θ's will be 0 ⇒ feature elimination

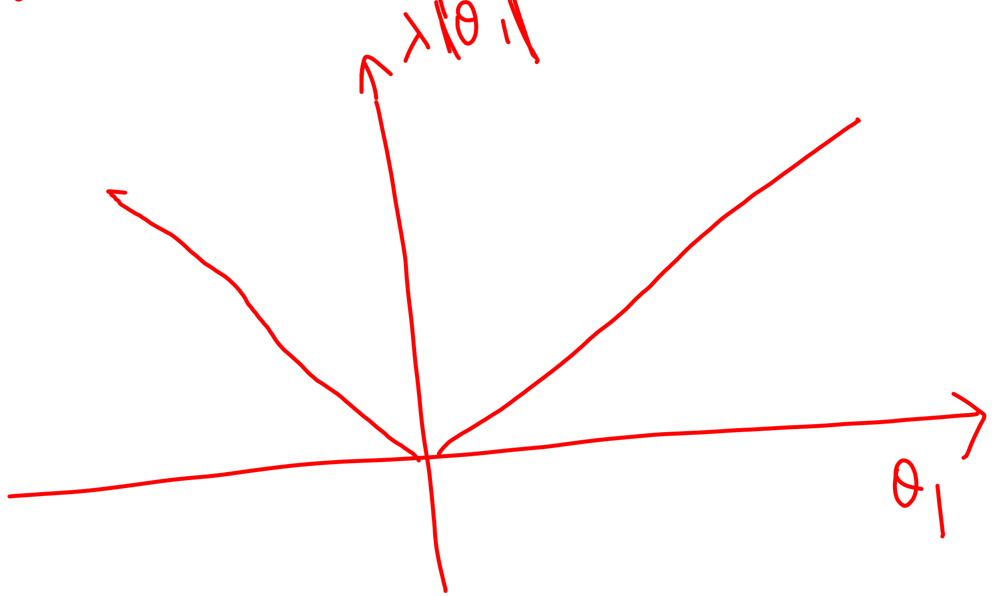
$$\theta_4 = 0, \quad \theta_8 = \theta_{20} = \theta_{50} = \theta_{70} = \theta_{82} = 0$$

$X_4, X_8, X_{20}, X_{50}, X_{70}, X_{82}$ ⇒ unused features in the model

⇒ features are automatically eliminated

Cost Function: Total Error + $\lambda\left(|\theta_1| + |\theta_2| + \dots + |\theta_{100}|\right)$

$$\boxed{\lambda|\theta_1|}$$



L2 Regularization

$= $ Ridge Regression

$R(\theta) = \sum_{i=1}^{n} |\theta_i|^{2}$

$\theta_0 \Rightarrow$ not included

$\sum |\theta_i|^3 \Rightarrow L3$

- L2 regularization adds a penalty equal to the square of the magnitude of coefficients
- This type of regularization does NOT result in sparse models with fewer coefficients
- Features are not eliminated from the model

- Multiple linear regression cost function with L2 regularization: **(Ridge Regression)**

$$J(\theta) = \underbrace{\frac{1}{2M} \sum_{i=1}^{M} (\theta_0 + \theta_1 X_{i1} + \theta_2 X_{i2} - Y_i)^2}_{\text{Mean Squared Error}} + \lambda \sum_{j=1}^{n} \theta_j^2$$

$+ \lambda (\theta_1^2 + \theta_2^2)$

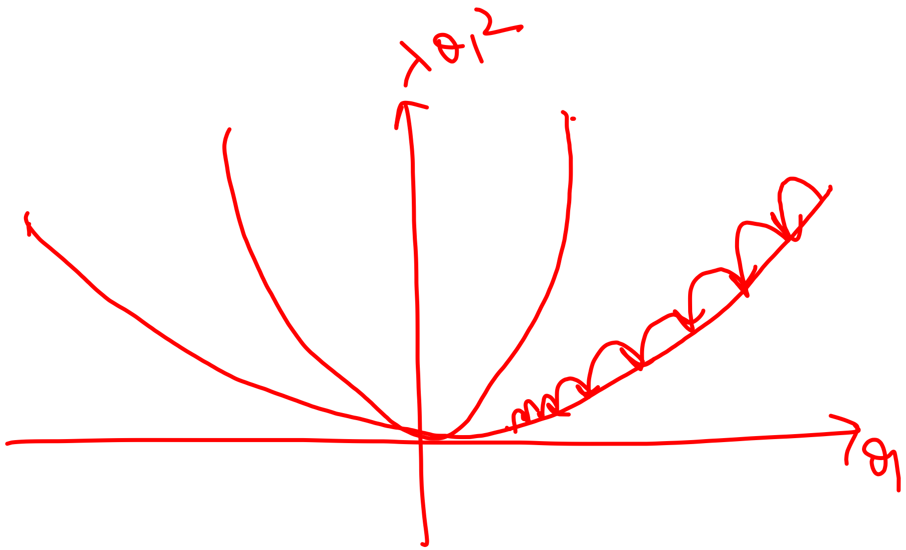
$\frac{1}{2} \frac{\partial x^2}{\partial x} = 2x$

θ_j^2 represents squared value of coefficient

In this case n will be 2 as we have considered 2 features

Note: Regularization is typically used when there are large number of features.

$$\text{Cost Function} = \text{Total Error} + \lambda(\theta_1^2 + \theta_2^2)$$



$$\theta_i \Leftarrow \theta_i - \eta \frac{\partial(\text{Cost Function})}{\partial\theta_i}$$

$$\theta_i \Leftarrow \theta_i - \eta \left[\frac{\partial(\text{Error})}{\partial\theta_i} + \frac{\partial(\lambda\theta_i^2)}{\partial\theta_i} \right]$$

$$\theta_i \Leftarrow \theta_i - \eta \left[\frac{\partial(\text{Error})}{\partial\theta_i} + 2\lambda\theta_i \right]$$

Assume: θ_i is not useful

$$\frac{\partial(\text{Error})}{\partial\theta_i} \approx 0$$

$$\theta_i \sim\Leftarrow \theta_i \underbrace{- 2\eta\lambda\theta_i}_{\text{updation}}$$

Key points regularization

Regularization can be applied to regression as well as classification models

- Regularization is useful when there are large number of features
- Regularization helps in avoiding overfitting
- L1 Regularization can be used for feature selection
- Regularization is useful in presence of multicollinear features

hyper parameter

Significance of λ:

- Very High value of λ results in higher penalty – less complex model (underfitting)
- Very low value of λ results in very low penalty (chance of overfitting – no benefit of regularization)

Questions?